



HGST Device Manager User Guide Version 3.1



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<i>Light Blue Italic</i>	<i>Text</i>	Used to indicate a hyperlink or “jump” to a related topic or subtopic. In addition, the text may be bold .

Revision History

Revision	Date	Page(s)	Description
-300	12/20/2014	All	Initial release.
-301	03/02/2015	All	Preliminary release.
-302	03/20/2015	19	Topic; Shared Library (Linux Installation) struck from user guide. Obsolete specification.
		20	Topic; Shared Library (Windows Installation) struck from user guide. Obsolete specification. Also added note explaining the necessity of using the HGST In-house Driver.
		21	Topic; Device References. Note 3 struck due to inaccuracy.
		23	Topic; Output Types. Minor edit of Sentence #2 to include <code>mini</code> (MINI) output type.
		29	Topic; <code>alias</code> command updated.
		15	Topic; Beta Limitations. Statement updated to reflect latest beta release.
		31	Topic; <code>configure-smart options temperature and availablespare</code> documented.
		35	Topic; <code>generate-report</code> Examples updated to include both redirection and console output.
		37	Topic; <code>get-feature NVMe Feature --feature options</code> documented.
		40	Topic; <code>get-log-page NVMe --page</code> options documented.
		45	Topic; <code>get-statistics</code> Example syntax corrected.
		46, 47, 48	Topic; <code>get-statistics</code> category examples updated.
		53	Topic; <code>resize</code> command. Added table listing valid resize ranges for <code>resize</code> command.
-303	04/15/2015	All	Struck references to “Beta” in preparation for GA Release.
		All	Global Search/Replace; The default is “mini”.
		viii	Exhibit A; struck Public Domain component “murmur3 hash function” from Table 1. Obsolete specification.
		15	Topic; “Beta Limitations” changed to “GA Limitations” and note re-worded to reflect GA software release.
		19	Topic; Linux Installation. Installation Notes updated.
		20	Topic; Windows Installation. Installation Notes updated.
		22	Figure 1 updated to show <code>mini</code> output format.
		28-52	The example output, where applicable, is now the <code>mini</code> , rather than the <code>text</code> , output format.
		26	Table 5, Command Exit Status Codes matrix updated.

Revision	Date	Page(s)	Description
-303	(continued)	47	Topic; <code>monitor</code> Command. Cautionary note added in regards to the <code>--interval</code> and <code>--count</code> values.
		48	Topic; <code>reset-to-defaults</code> Command. Cautionary note added in regards to power cycle of system.
		49	Topic; <code>resize</code> Command. Cautionary notes added in regards to <code>resize</code> command functionality and support.
-304	07/17/2015	Title	Title page information updated.
		viii	Exhibit A; updated to reflect legal requirements.
		22	Topic; Linux Directory Trees vs. Windows Device Instance IDs. Informational only.
			Topic; Short Form Syntax. Table of “Long Form” and “Short Form” equivalent options implemented in HDM 3.1.
			Topic; Configuration Files. The short form <code>-c</code> option documented for the <code>--config PATH</code> option.
		24	Topic; Output Types. The short form <code>-o</code> option documented for the <code>--output-format FORMAT</code> option.
		29	Topic; Command Confirmation developed for destructive commands.
		34	Topic; <code>configure-trace</code> . The short form <code>-f</code> option documented for the <code>--file FILE</code> parameter.
		54-56	Topic; <code>manage-namespaces</code> developed for HDM 3.1.
		57	Topic; <code>manage-power</code> developed for HDM 3.1.
		60	Topic; <code>prepare-for-removal</code> developed for HDM 3.1.
		65	Topic; <code>secure-purge</code> developed for HDM 3.1.
305	07/30/2015	32	Topic; <code>configure-smart</code> command. Note added to clarify available spare threshold limitations vs. currently available spare threshold value and read-only mode.
		57	Topic; <code>manage-power</code> command. Note added in regards to proper usage of command and the possible adverse consequences to changing the power state of the device.
		62	Topic; <code>resize</code> command. Updated table listing the minimum and maximum ranges according to drive capacity.

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Scope

Overview

The HGST Device Manager (HDM) is designed to efficiently assist in the administration of Solid-State Drive (SSD) and Solid-State Accelerator (SSA) devices. While many IT organizations have embraced the benefits of solid-state technology, many of the current hard disk drive utilities have proven woefully inadequate in the management of enterprise solid-state devices. HDM provides the ability to easily update and configure SSDs and SSAs.

Audience

This user guide is intended for system administrators, network administrators, and other IT professionals. It is therefore written specifically for a technically advanced audience; it is not intended for end-users that will eventually purchase the commercially available product. The *user*, as referenced throughout this guide, is primarily concerned with industrial, commercial and military networking applications.

Features

Feature	Description
Cross-Platform Solution	Simplifies the management of SSDs and SSAs in Linux 64-Bit and Windows 64-bit environments.
Command Line Interface (CLI)	A consistent command line interface that supports the use of Linux scripts or Windows batch files to automate tasks. The user can choose from four (4) modes of output: <code>mini</code> , <code>text</code> , <code>json</code> and <code>csv</code> . The <code>mini</code> option (default) presents the output in an abridged format, the <code>text</code> output is a human-readable format, the <code>json</code> output is for a scripting environment, and the <code>csv</code> format can be used for data mining purposes and analysis.

Specifications

Specification	Description
Overview	Software solution that operates independently of host hardware for managing multiple drives.
Solid-State Device Compatibility	See Supported Products .
Operating Systems	See Operating Systems under Prerequisites .
Capabilities	All inclusive solid-state device solution; firmware management, namespace management, power management, report generation, secure erase and secure purge, factory reset and format, resize, statistics, diagnostic capture, and S.M.A.R.T. data retrieval.
Management	CLI utilities to manage solid-state devices.

Supported Products

HDM can update and configure the following SSDs as outlined in Table 1.



HDM 3.1 requires that Firmware Version KMGNP110 or later be installed on the SN100 or SN150 Series SSDs; otherwise, the execution of the `resize`, `reset-to-defaults` and `manage-namespaces` commands will return an error. Please see the Release Notes for more information.

Table 1: Supported Products

Product Family	Description
Ultrastar® SN100 Series	SN100 NVMe PCIe 3.0 x4 Lane 2.5-Inch SFF Solid-State Drives
Ultrastar® SN150 Series	SN150 NVMe PCIe 3.0 x4 Lane HH-HL Card Solid-State Drives

Installation

Overview

This section addresses issues regarding the compatibility, system requirements, installation and configuration of HDM.

Prerequisites



The prerequisite platform or software described below should be installed and configured before you install HDM.

Operating Systems



HDM 3.1 only supports the kernels that are supplied with the operating systems distributed by the OS vendors. If the user should compile any other kernel versions into the operating systems, then the configuration shall not be considered officially supported by HDM.

Operating System	Description
Red Hat Enterprise Linux	Red Hat Enterprise Linux (RHEL) 6, 64-bit with supplied kernels. Red Hat Enterprise Linux (RHEL) 7, 64-bit with supplied kernels.
CentOS	Community Enterprise Linux (CentOS) 6, 64-bit with supplied kernels. Community Enterprise Linux (CentOS) 7, 64-bit with supplied kernels.
SUSE Linux	SUSE Linux Enterprise Server (SLES) 11, 64-bit with supplied kernels. SUSE Linux Enterprise Server (SLES) 12, 64-bit with supplied kernels.
Microsoft Windows	Windows Server 2008 R2, 64-bit Windows Server 2012, 64-bit Windows Server 2012 R2, 64-bit Windows 7, 64-bit Windows 8.1, 64-bit

User Privileges

Prerequisite	Description
Linux Distributions	Root
Microsoft Windows	Administrator

Supported Drivers



The minimum version of the HGST NVMe driver required by HDM 3.1 is 1.0.1.50625. The corresponding driver file names will be prefixed by "HGST-NVMe-10150625".

Prerequisite	Description
HGST NVMe Windows Driver	Windows NVMe Driver provided by HGST, Inc. Note: HDM is not compatible with Windows In-box drivers due to the limitations of these drivers.
HGST NVMe Linux Driver	Linux NVMe Driver provided by HGST, Inc.
Linux In-box Drivers	Drivers included with Linux Distributions.

System Hardware Requirements

Prerequisite	Description
CPU	Intel Single-Core 64-bit Processor AMD x86_64 Processor
Display	Super VGA (1024 x 768) or higher resolution monitor.
Hard Disk Drive	2.0 MB available disk space minimum.
Memory	512MB RAM
Network	Internet connection required for downloads and upgrades.
Optical Drive	CD-ROM/DVD-ROM optional.
Peripherals	Keyboard and Mouse (Printer optional).

Installation Packages

The HDM interface is encapsulated within installation packages for the Linux Distributions and Microsoft Windows. An authorized representative will e-mail the desired version of HDM to the user. The packages must be unpacked in a default directory or a temporary directory. It is recommended that the user create the default or temporary directories off the root of the drive hierarchy. Table 2 lists the available installer packages and descriptions.

Table 2: HDM Archive Descriptions

HGST Part Number	Archive Name	Description
66000-00053-002	HGST Device Manager-3.1.0-win64.msi	Windows 64-Bit; HDM CLI
66000-00054-002	hdm-core-3.1.0-1-x86_64.rpm	Linux 64-Bit; HDM CLI

Event Logging

HDM supports the use of event logging as a method of tracking drive health and operation. Event messages are usually logged to an operating system event log such as the Event Viewer (Windows) or /var/log/syslog (Linux).

Installation



It is strongly recommended that the user completely review each installation procedure respective to the individual platform, including any Installation Notes.



It is strongly recommended that the user uninstall any previous versions of HDM before installing the latest version.

Linux Installation

To install HDM under Linux:

1. Logon to the system with `root` privileges.
2. Create a temporary installation folder or directory.
 1. Download the HDM 3.1 rpm file: `hdm-core-3.1.0-1-x86_64.rpm` to the temporary installation folder or directory.

Note: The installer is only compatible with 64-bit systems.
 2. Open a terminal window and change directories to the temporary installation folder or directory.
 3. Run the following command: `rpm -ivh hdm-core-3.1.0-1-x86_64.rpm`

Note: The installer will check whether the necessary prerequisite packages are present before continuing the installation.

Installation Notes

You will notice that after HDM is installed:

1. Several HDM-related files are contained within the `/opt/HGST/hdm/` directory.
2. The HDM CLI can be launched via the `/opt/HGST/hdm/hdm` binary.
3. Running the command of `rpm -qa | grep hdm` should return the HDM packages that were installed.

Uninstall Procedure

To uninstall HDM:

1. Logon to the system with `root` privileges.
2. HDM CLI Uninstall:
 1. Open a terminal window or console.
 2. Run `rpm -e <hdm-core>`.

Windows Installation



The user must install the HGST In-house Driver before using HDM to manage HGST devices under Windows.

To install HDM under Windows:

1. Logon to the system with Administrator privileges.
2. Create a temporary installation folder or directory.
3. Download the HDM 3.1 msi file: HGST Device Manager-3.1.0-win64.msi to the temporary installation folder or directory.

Note: The installer is only compatible with 64-bit systems.

4. Double-click the **Installer Package (msi)**.

Notes:

1. If a Microsoft SmartScreen filter message appears informing you that the package being installed cannot be verified, click **More Information** and then allow the package to be installed.
2. Enter any administrative credentials if so prompted.
5. The HGST Device Manager Setup Window appears.
6. Click **Next** to continue. The End-User License Agreement will appear.
7. Review the terms of the EULA and then place a check mark in the **I accept the terms in the License Agreement** check box.
8. Click **Next** to continue. You are prompted to choose the location of the working directory. The default path will appear in the text box, but you can use the **Change...** option to navigate to an alternate installation directory.
9. Click **Next** to continue. You are now prompted to install HGST Device Manager.
10. Click **Install**. The HDM CLI will be installed.

Installation Notes

You will notice that after HDM is installed:

1. All the HDM-related files are contained in the installation path that was specified in the Setup Wizard.
2. The **Add/Remove Programs** panel or **Programs and Features** panel under the **Control Panel** will contain the application of “HGST Device Manager”.
3. All command execution for the CLI requires the use of a terminal / command window and administrator privileges.

Uninstall Procedure

To uninstall HDM:

1. Open the Control Panel window.
 - a. Navigate to the Add/Remove Programs panel, or;
 - b. Navigate to the Programs and Features panel.
2. Click the application “HGST Device Manager” or “HDM” in the list to select it.
3. Click the Uninstall button.
4. Confirm the removal of HDM.

Command Line Interface

Overview

This section explains the usage and capabilities of the CLI (Command Line Interface) and provides basic and advanced device management and diagnostic functions. The CLI can also be used in situations where automated tasks can be executed as scripts or batch files.

Command Execution

The syntax for command execution is consistent across the various platforms. In this section, the commands are presented in the platform neutral form of **hdm**. The user should have a practical knowledge of navigating the command line interface for the specific system platform.

Command Syntax

The commands and options use the same syntax across the platforms. The spaces or delimiters are taken literally, while the brackets are ignored:

```
hdm <command> <command args>
```

Where **hdm** is a “meta” command that invokes the various management or administrative commands, while **<command>** is the actual operation.

Command Arguments

The **<command args>** (command arguments) may consist zero or multiple combinations of options or flags. An option is considered an input parameter that requires a user supplied value, while flags modify the command behavior.

Device References

A *device-reference* describes how a device is accessed when executing a command and may be expressed as shown in Table 3. See [Linux Directory Trees vs. Windows Device Instance IDs](#).

Table 3: Device References

Argument	Description	Example
--uid (-u)	The unique identifier (UID) of the device.	--uid 1C58STM0001A6758HUSPR3216AHP301
--path (-p)	The OS native path or device instance ID.	--path /dev/nvme0
--alias (-a)	The HDM internal alias.	--alias myssd01

Notes:

1. The unique identifier is device specific and HDM uses the following unique identifiers:
 - a. NVMe Controllers: A concatenation of the PCI Vendor ID, Serial Number, Model Number (Identify Controller Structure).
 - b. NVMe Namespaces: IEEE Extended Unique Identifier (EUI64) (Identify Namespace Structure)
2. While the most intuitive mechanism may be to reference a device by using the OS native path, the enumeration of the devices, the controller paths, and namespaces may change after a reboot.

Device Reference Examples

For example, the user may want to obtain the properties of a device using the `get-info` command. The user must therefore specify the *device-reference*; the OS path, an alias or a device unique identifier.

OS Path Example

In the following example, the *<device-reference>* argument is the **OS path** and follows the `--path` input option:

```
hdm get-info --path /dev/nvme0
```

Alias Example

In the following example, the *<device-reference>* argument is the **alias** that was assigned to the device and follows the `--alias` input option:

```
hdm get-info --alias myssd01
```

Device Unique Identifier Example

In the following example, the *<device-reference>* argument is the **device unique identifier** assigned to the device and follows the `--uid` input option:

```
hdm get-info --uid 1C58STM001A6758HUSPR3216AHP301
```

Linux Directory Trees vs. Windows Device Instance IDs

It is important to understand that Linux and Windows manage devices using different methods. Linux implements the `/dev/` directory tree scheme; the `/dev/` directory tree contains all the “device” files, allowing the user to perform operations on a specific device.

For example, if the user executes the `hdm scan` command on a Linux system, then the `/dev/` directory tree scheme for devices is used, i.e., “Device Path = `/dev/nvme0`.”

However, under Windows, HDM uses the device instance paths or device instance IDs assigned by the system. A device instance ID is a unique identification string assigned by the Plug and Play (PnP) Manager to each device node in the device tree. The format of the string consists of an instance ID concatenated to a device ID, i.e., `<device-ID>\<instance-specific-ID>`. The device instance ID is persistent across system restarts.

If the user executes the `hdm scan` command on a Windows system, then the device instance string is used, i.e., “Device Path = `\\?\pci#ven_1c58&dev_0003&subsys_00031c58&rev_05#4&11cc8299&0&0009#{2accfe60-c130-11d2-b082-00a0c91efb8b}`.”

Short Form Syntax

Table 4 lists the “short form” syntax that is available for the following <command args> (command arguments).

Table 4: Short Form Command Arguments

Long Form	Short Form	Example
<code>--output-format FORMAT</code>	<code>-o</code>	<code>hdm get-state -o mini -path /dev/nvme0</code>
<code>--config PATH</code>	<code>-c</code>	<code>hdm get-state -c \$HOME/hdm.ini --path /dev/nvme0</code>
<code>--path DEV_PATH</code>	<code>-p</code>	<code>hdm get-state -p /dev/nvme0</code>
<code>--uid UNIQUE_ID</code>	<code>-u</code>	<code>hdm get-state -u 1C58STM0001A6758HUSPR3216AHP301</code>
<code>--alias ALIAS</code>	<code>-a</code>	<code>hdm get-state -a myssd01</code>
<code>--file FILENAME</code>	<code>-f</code>	<code>hdm configure-trace -f nvme0trace --level 3</code>
<code>--interval INTERVAL</code>	<code>-i</code>	<code>hdm get-statistics -i 14 --category performance --path /dev/nvme0</code>

Configuration Files

The `--config PATH` or `-c` option allows the user to specify a configuration file other than the default. The option expects the user to specify the path to the configuration file, `hdm.ini`. The configuration file is usually located in the `$HOME/.hdm` directory by default. If the user does not specify `$HOME`, then the current directory is used.



The `--config PATH` option is for advanced usage and is not recommended for normal situations

Listing the Installed Devices

The first command that the user should run is **scan**. The **scan** command will list all the names of the drives and devices installed in the system and that are currently recognized by the operating system. These names can then be used in the execution of subsequent commands. For example, **get-info** can use either the `{--path PATH |--uid UID |--alias ALIAS}` options to extract drive information.

```
hdm scan
[1C58STM0001A6758HUSPR3216AHP301]
  Device Type      = Ultrastar SN150
  Device Path      = /dev/nvme0
  UID              = 1C58STM0001A6758HUSPR3216AHP301
  Vendor Id        = 1C58
  Device Id        = 0003
  Vendor Name       = HGST
  Model Name       = HUSPR3216AHP301
[000CCA0060032E80]
  Device Type      = NVMe Namespace
  Device Path      = /dev/nvme0n1
  UID              = 000CCA0060032E80
  Parent UID       = 1C58STM0001A6758HUSPR3216AHP301
  Parent Device Path = /dev/nvme0
  Namespace ID     = 1
Results for scan: Operation succeeded on 2 of 2 devices.
```

Figure 1: Listing the Installed Devices

Output Types

The `--output-format FORMAT` or `-o` option allows for flexibility of output after successful command execution. The user substitutes `mini` (for MINI output), `text` (for ASCII text output), `json` (for JSON output), or `csv` (for Comma-Separated Values output) for the `FORMAT` value.

MINI Output

The MINI output is an abridged, simplified format of the command output. The use of `mini` is optional; the format is output by default if the user does not specify the `text`, `json`, or `csv` formats. For example:

```
hdm get-state --output-format mini --path /dev/nvme0
hdm get-state -o mini --path /dev/nvme0

OR

hdm get-state --path /dev/nvme0
```

The output will be in the `mini` output format:

```
[/dev/nvme0]
  Device Type      = Ultrastar SN150
  Device Path      = /dev/nvme0
  UID              = 1C58STM0001A82F8HUSPR3216AHP301
  Device Status    = Ready
  Power Consumption = 6.917188
  Life Gauge       = 100
  Dual Port Mode   = Single Port
  Encryption Mode  = No Encryption
Results for get-state: Operation succeeded.
```

Figure 2: MINI Output Example

Text Output

The user can specify the `text` option to specify that the output appear on the computer screen as shown in the following figure. The output is presented in a tabular, human-readable format.

```
hdm get-state --output-format text --path /dev/nvme0
```

```
=====
                        Results for get-state
=====
+-----+-----+
| Device Type           | Ultrastar SN150           |
+-----+-----+
| Device Path           | /dev/nvme0                 |
+-----+-----+
| UID                   | 1C58STM0001A82F8HUSPR3216AHP301 |
+-----+-----+
| Device Status         | Ready                      |
+-----+-----+
| Power Consumption     | 6.925375                   |
+-----+-----+
| Life Gauge            | 100                        |
+-----+-----+
| Dual Port Mode        | Single Port                 |
+-----+-----+
| Encryption Mode       | No Encryption              |
+-----+-----+
                        Result: Operation succeeded.
=====
```

Figure 3: Text Output Example

JSON Output

JSON is an acronym for JavaScript Object Notation and is an open standard format that uses human-readable text to transmit data objects consisting of attribute-value pairs. The format is often employed for a scripting environment. To specify JSON output, use the `--output-format json` option.

```
hdm get-state --output-format json --path /dev/nvme0
```

```
{
  "status":0,
  "results":[
    {
      "device_type":"Ultrastar SN150",
      "device_path":"/dev/nvme0",
      "uid":"1C58STM0001A82F8HUSPR3216AHP301",
      "device_status":"0",
      "power_consumption":7.024563,
      "life_gauge":100,
      "dual_port_mode":"Single Port",
      "encryption_mode":"No Encryption",
      "status":0
    }
  ]
}
```

Figure 4: JSON Output Example

CSV Output

CSV is an acronym for Comma-Separated Values (a.k.a. Character-Separated Values). The CSV file type is a compilation of tabular data (numbers/text) in a plain-text form; the file is a sequence of characters, with no interpretation of binary numbers, divided into records that are subdivided into fields. These fields are separated by a literal comma (.). The user should specify the filename extension of *.csv to easily identify the file.

To specify CSV output, use the `--output-format csv` option. Figure 5 shows an example of CSV output within a text editor. It is then possible to import the CSV file into a spreadsheet program, or use the CSV data contained in the file for data mining purposes or analysis.

```
hdm get-smart --output-format csv --path /dev/nvme0

DEVICE_REFERENCE,STATUS,DEVICE_TYPE,DEVICE_PATH,UID,ALIAS,READ_ERROR_RATE
,WRITE_ERROR_RATE,ERASE_ERROR_RATE,VERIFY_ERROR_RATE,DATA_UNITS_READ,DATA
_UNITS_WRITTEN,HOST_READ_COMMANDS,HOST_WRITE_COMMANDS,AVAILABLE_SPARE,AVA
ILABLE_SPARE_THRESHOLD,AVAILABLE_SPARE_CRITICAL_WARNING,PERCENTAGE_USED,A
VAILABLE_RESERVED_SPACE,TEMPERATURE,TEMPERATURE_THRESHOLD,TEMPERATURE_CRI
TICAL_WARNING,REFERENCE_TEMPERATURE,MAXIMUM_TEMPERATURE,MINIMUM_TEMPERA
TURE,TOTAL_TIME_OVER_REFERENCE_TEMPERATURE,POWER_ON_HOUR_WHEN_AT_MAX_TEMP
ERATURE,POWER_ON_HOUR_WHEN_AT_MIN_TEMPERATURE,DEVICE_RELIABILITY_CRITICAL
_WARNING,MEDIA_READ_ONLY_MODE_CRITICAL_WARNING,VOLATILE_MEMORY_BACKUP_DEV
ICE_FAILURE_CRITICAL_WARNING,CONTROLLER_BUSY_TIME,POWER_CYCLES,POWER_ON_H
OURS,UNSAFE_SHUTDOWNS,MEDIA_AND_DATA_INTEGRITY_ERRORS,NUMBER_OF_ERROR_INF
ORMATION_LOG_ENTRIES
/dev/nvme0,Success,Ultrastar
SN150,/dev/nvme0,1C58STM0001A82F8HUSPR3216AHP301,,4.863834,0.000000,0.000
000,0.000000,55557326,115373711,3019544347,5609101100,100,10,false,0,,45,
90,false,90,73,24,0,2858,7,false,false,false,1855,2,215,1,0,1
```

Figure 5: CSV Output Example

Commands

Table 5 lists the valid commands for HDM 3.1 that must be specified prior to an input or value parameter.



While HDM commands and options are usually not case-sensitive, the user should be aware that paths, UIDs, aliases, etc. are case sensitive.

Table 5: HDM Commands

Command	Description
alias	Sets or clears a user supplied name for a device.
capture-diagnostics	Captures diagnostic data from a physical device.
configure-smart	Sets or clears SMART thresholds on a device.
configure-trace	Configures the trace level and destination file.
format	Performs a low-level format on devices.
generate-report	Retrieves information and statistics for devices.
get-feature	Retrieves feature data from devices.
get-info	Retrieves properties from devices.
get-log-page	Retrieves log pages from devices.
get-smart	Retrieves SMART statistics and thresholds from devices.
get-state	Retrieves state information from devices.
get-statistics	Retrieves statistics from devices.
help	Prints command help information.
locate	Enables or disables the LED beacon for physical devices.
manage-firmware	Displays and updates firmware on physical devices.
manage-namespaces	Displays and manages namespaces on a capable physical device.
manage-power	Displays and controls power states of a physical device.
monitor	Continuously retrieves device information at a specified interval.
prepare-for-removal	Prepares physical devices for safe removal from the system.
reset-to-defaults	Resets physical devices to factory default configuration.
resize	Changes the user capacity of physical devices.
scan	Lists the storage devices attached to the host.
secure-erase	Securely erases all user data from physical devices.
secure-purge	Performs a secure purge on physical devices.
version	Displays the HDM version and build information.

Help System

Complete Command Listing

The `help` command provides assistance for all supported HDM commands. If `help` is used without any arguments, it will list the available commands as shown below.

```
usage: hdm <command> [--output-format FORMAT] [--config PATH] <command args>
```

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.

COMMANDS:

<code>alias</code>	Sets or clears a user supplied name for a device.
<code>capture-diagnostics</code>	Captures diagnostic data from a physical device.
<code>configure-smart</code>	Sets or clears SMART thresholds on physical devices.
<code>configure-trace</code>	Configures the trace level and destination file.
<code>format</code>	Performs a low-level format on devices.
<code>generate-report</code>	Retrieves information and statistics for devices.
<code>get-feature</code>	Retrieves feature data from devices.
<code>get-info</code>	Retrieves properties from devices.
<code>get-log-page</code>	Retrieves log pages from devices.
<code>get-smart</code>	Retrieves SMART statistics and thresholds from devices.
<code>get-state</code>	Retrieves state information from devices.
<code>get-statistics</code>	Retrieves statistics from devices.
<code>help</code>	Prints command help information.
<code>locate</code>	Enables or disables the LED beacon for physical devices.
<code>manage-firmware</code>	Displays and updates firmware on physical devices.
<code>manage-namespaces</code>	Displays and manages namespaces on a capable physical device.
<code>manage-power</code>	Displays and controls power states of a physical device.
<code>monitor</code>	Continuously retrieves device information at a specified interval.
<code>prepare-for-removal</code>	Prepares physical devices for safe removal from the system.
<code>reset-to-defaults</code>	Resets physical devices to factory default configuration.
<code>resize</code>	Changes the user capacity of physical devices.
<code>scan</code>	Lists the storage devices attached to this host.
<code>secure-erase</code>	Securely erases all user data from physical devices.
<code>secure-purge</code>	Performs a secure purge on physical devices.
<code>version</code>	Display HDM build and version information.

Run '`hdm help <command>`' for more information about a particular `hdm` command.

For more information about HDM, refer to the HDM User Guide and Release Notes documents.

Specific Command Help

If the `<command>` follows the `help` command, e.g., `hdm help scan`, it will list only the options that are relevant to the `scan` command:

```
usage: hdm scan          [--output-format FORMAT] [--config PATH]
```

List the storage devices attached to this host.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.

Command Confirmation

HDM commands that can perform potentially destructive operations that result in the deletion of user and system data now require user confirmation. The `format`, `manage-namespaces`, `reset-to-defaults`, `resize`, `secure-erase`, and `secure-purge` commands will now prompt for confirmation. The user may override the prompt using the `--force` option. The following confirmation prompt will appear for destructive commands:

Press 'y' to continue

Press 'y' to confirm. The user can press any other key to cancel the operation.

Command Exit Status Codes

HDM may return one or more exit status codes to indicate the outcome of an executed command. The following table lists the possible status results that may occur.

Table 6: Command Exit Status Codes

Exit Status	Description
0	Success
1	Invalid user supplied parameter or value.
2	Device error.
3	I/O error.
4	Permissions error.
5	System error.
6	Qualified success.
7	Internal error.

alias

The `alias` command allows a user to assign or remove a user-readable name, or a name supplied by the user to a drive or namespace, as a substitute for the system assigned name. The [device-reference](#) must refer to a single physical device. The user can then specify the use of the alias (`--alias ALIAS`) in all subsequent HDM commands that require a [device-reference](#) parameter. Using the `--clear-alias` flag will remove the alias name from the [device-reference](#).

Synopsis

```
usage: hdm alias                [--output-format FORMAT] [--config PATH]
                                [--path PATH |--uid UID |--alias ALIAS}
                                [--name NAME] [--clear-alias]
```

Sets or clears a user supplied name for a device.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--name NAME</code>	The name to be used for this operation.
<code>--clear-alias</code>	Clear the alias for the specified device.

Examples

To assign an alias to a device:

```
hdm alias --name alias-name --path /dev/nvme0
```

OR

```
hdm alias --name nvme32Tn1 --path /dev/nvme0
```

To remove an alias from a device:

```
hdm alias --name nvme32Tn1 --clear-alias --path /dev/nvme0
```

To assign an alias to a namespace:

```
hdm alias --name alias-name --path /dev/nvme0n1
```

OR

```
hdm alias --name nvme32Tn1 --path /dev/nvme0n1
```

To remove an alias from a namespace:

```
hdm alias --name nvme32Tn1 --clear-alias --path /dev/nvme0n1
```

capture-diagnostics

The `capture-diagnostics` command will gather diagnostic data for a device and compile it to a specific HGST container file that the user can then send to technical support for analysis. The contents of the captured data vary by drive type. The [device-reference](#) must refer to a single physical device. Namespaces are not supported.

Synopsis

```
usage: hdm capture-diagnostics  [--output-format FORMAT] [--config PATH]
                                [--path PATH |--uid UID |--alias ALIAS]
                                [--file FILE]
```

Captures diagnostic data from a physical device.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--file FILE</code>	File name or path.

Example

```
hdm capture-diagnostics --file nvme0diags --path /dev/nvme0
```

configure-smart

The `configure-smart` command will set and/or clear the SMART `temperature` or `availablespace` thresholds. The `--set` and `--clear` options may be specified only once on a single command and combinations of `--set` and `--clear`, and `--clear all` are not allowed in the same command invocation. The `-clear-all` flag will reset all of the available thresholds for the device.



Ultrastar NVMe PCIe Devices Only: Users should be aware that if the `availablespace` threshold is exceeded, the device will enter read-only mode. Users should be careful not to specify too high a value for the available spare threshold, or that the value be greater than the currently available spare value.

Synopsis

```
usage: hdm configure-smart      [--output-format FORMAT] [--config PATH]
                                [--path PATH |--uid UID |--alias ALIAS]
                                [--set] [--clear-all] [--clear]
```

Sets or clears SMART thresholds on physical devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--set</code>	Set the value of a specific SMART threshold.
<code>--clear-all</code>	Clear all threshold warnings and change all SMART thresholds for the specified devices to default values.
<code>--clear</code>	Clear all threshold warnings and change a specific SMART threshold for the specified devices to its default value.

Options

--set Value	Description
temperature=(Celsius)	The <code>temperature</code> value is used to either set or clear the SMART temperature threshold. The threshold is expressed in degrees Celsius. For example, to set the temperature threshold at 60°C, the expression would be <code>--set temperature=60</code>. The user can reset the temperature threshold to the factory default using <code>--clear temperature</code> and omitting the Celsius value.
availablespace=(percentage)	The <code>availablespace</code> value is used to either set or clear the available spare threshold. The threshold is expressed as a percentage, e.g., to set the threshold at 10%, <code>--set availablespace 10</code>. The user can reset the available threshold to the factory default using <code>--clear availablespace</code> and omitting the percentage value.

Examples

To set the SMART temperature threshold:

```
hdm configure-smart --set temperature=60 --path /dev/nvme0
```

To clear the SMART temperature threshold:

```
hdm configure-smart --clear temperature --path /dev/nvme0
```

To set the SMART availablespace threshold:

```
hdm configure-smart --set availablespace=20 --path /dev/nvme0
```

To clear the SMART availablespace threshold:

```
hdm configure-smart --clear availablespace --path /dev/nvme0
```

To clear the temperature and availablespace SMART thresholds:

```
hdm configure-smart -clear-all --path /dev/nvme0
```

Note: The following is an example of an *invalid* configure-smart command, with the --set and --clear options being used in the same command line:

```
hdm configure-smart --set temperature=60 --clear availablespace  
--path /dev/nvme0
```

configure-trace

The `configure-trace` command allows the user to specify the path and filename for the trace data and the trace level. The `--level` parameter determines the message class (error, warning) to record in the trace file. The trace file specified by the `-file` or `-f` parameter will create the filename in the specified path if it does not exist. If the file name currently exists then the trace data will be appended to the file. The messages in the trace file are generated by HDM and record a variety of error conditions, including the error return codes from the OS, NVMe status codes from the drive, and internal HDM errors.

Synopsis

```
usage: hdm configure-trace  [--output-format FORMAT] [--config PATH]
                             [--file FILE] [--level LEVEL]
```

Configures the trace level and destination file.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--file FILE</code>	File name or path
<code>--level LEVEL</code>	Trace level to be used for tracing. 0 - do not trace any messages 1 - includes only error-class messages 2 - includes error-class and warning-class messages 3 - includes all trace messages

Options

Option	Value	Description
<code>--file FILE</code>	path/trace filename	Specify the path and filename for the trace data. The file will be compiled in the <code>\$HOME/.hdm</code> directory if the path is not specified.
<code>--level LEVEL</code>	0, 1, 2, 3	The <code>LEVEL</code> value specifies the trace level to be used for tracing: 0 = Do not trace any messages. 1 = Trace only error-class messages. 2 = Trace error-class and warning-class messages. 3 = Trace all messages.

Example

```
hdm configure-trace --file nvme0trace --level 2
```

format

The `format` command will perform a low-level format on the targeted device or namespace using the specified sector size and DIF (Data Integrity Field) level. The default configuration will be used if the sector size, metadata size and/or the DIF level are not specified. The sector size, metadata size, and DIF level are required parameters when targeting a device configured with multiple namespaces. If the command is used for an NVMe device, then the format will be applied to all NVMe namespaces on the device. The user will be prompted to confirm unless the `--force` option is specified.

Synopsis

```
usage: hdm format                [--output-format FORMAT] [--config PATH]
                                [--path PATH |--uid UID |--alias ALIAS]
                                [--sector-size SIZE] [--dif-level LEVEL]
                                [--metadata-size SIZE] [--force]
```

Performs a low-level format on devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--sector-size SIZE</code>	Size of each sector in bytes.
<code>--dif-level LEVEL</code>	Level of DIF protection.
<code>--metadata-size SIZE</code>	Size of metadata in bytes.
<code>--force</code>	Bypass all confirmation messages.

Options

Option	Value	Description
<code>--sector-size SIZE</code>	512, 4096	The <code>SIZE</code> value specifies the new size of sector on the device or namespace.
<code>--metadata-size SIZE</code>	0, 8	The <code>SIZE</code> value specifies the new size for the metadata that includes information about the sector and the protection information. A <code>SIZE</code> of 0 is only valid if no protection level is specified; a <code>SIZE</code> of 8 is necessary when formatting the device with a valid DIF (protection) level.
<code>--dif-level LEVEL</code>	0, 1, 2, 3	The <code>LEVEL</code> value specifies the DIF (protection) level. 0 = No Protection. 1 = Type 1 protection. 2 = Type 2 protection. 3 = Type 3 protection.

Examples

To perform a default format of a device:

```
hdm format --path /dev/nvme0
```

To format a device with DIF protection:

```
hdm format --sector-size 512 --metadata-size 8 --dif-level 3 --path /dev/nvme0
```

To perform a default format of a namespace:

```
hdm format --path /dev/nvme0n1
```

To format a namespace with DIF protection:

```
hdm format --sector-size 512 --metadata-size 8 --dif-level 3 --path /dev/nvme0n1
```

generate-report

The `generate-report` command will compile device, state, statistics, and S.M.A.R.T. information for all the devices in the system or for a specific device. The report is a compilation of the `get-info`, `get-state`, `get-smart` and `get-statistics` commands. The user can redirect the output using the OS-specific `stdout` or `stderr` operators. The user must specify the [device-reference](#) if compiling a report for a single device; otherwise, the command will compile a report on all devices installed in the system.

Synopsis

```
usage: generate-report      [--output-format FORMAT] [--config PATH]
                             [--path PATH |--uid UID |--alias ALIAS]
```

Retrieves information and statistics for devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.

Examples

To generate a report for a specific device in the system and redirect the output to a text file:

```
hdm generate-report --path /dev/nvme0 > nvme0rpt.txt
```

To generate a report for all devices in the system and redirect the output to a text file:

```
hdm generate-report > nvmeallrpt.txt
```

OR

To generate a report for a specific device in the system and redirect the output to the console:

```
hdm generate-report --path /dev/nvme0
```

To generate a report for all devices in the system and redirect the output to the console:

```
hdm generate-report
```

get-feature

The `get-feature` command will retrieve and display a specific feature that is supported by the target device. The ID of the `FEATURE` is specified after the `--feature` option and can be expressed as a whole number or in hexadecimal (1 or 0x01).

Synopsis

```
usage: hdm get-feature      [--output-format FORMAT] [--config PATH]
                             [--path PATH | --uid UID | --alias ALIAS]
                             [--feature FEATURE]
```

Retrieves feature data from devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--feature FEATURE</code>	Identifier of the feature. This value must be an integer (e.g. 1) or a prefixed hexadecimal number (e.g. 0x01).

Options

--feature Value	Description
1 (0x01)	NVMe Feature 01h, Arbitration.
2 (0x02)	NVMe Feature 02h, Power Management.
3 (0x03)	NVMe Feature 03h, LBA Range Type; only supported for Ultrastar NVMe namespaces.
4 (0x04)	NVMe Feature 04h, Temperature Threshold.
5 (0x05)	NVMe Feature 05h, Error Recovery.
6 (0x06)	NVMe Feature 06h, Volatile Write Cache.
7 (0x07)	NVMe Feature 07h, Number of Queues.
8 (0x08)	NVMe Feature 08h, Interrupt Coalescing.
9 (0x09)	NVMe Feature 09h, Interrupt Vector Configuration; not supported by Ultrastar NVMe firmware (controller or namespace).
10 (0x0a)	NVMe Feature 0Ah, Write Atomicity.
11 (0x0b)	NVMe Feature 0Bh, Asynchronous Event Configuration.
12 (0x0c)	NVMe Feature 0Ch, Autonomous Power State Transition.
128 (0x80)	NVMe Feature 80h, Software Progress Marker.
129 (0x81)	NVMe Feature 81h, Host Identifier.
130 (0x82)	NVMe Feature 82h, Reservation Notification Mask; not supported by Ultrastar NVMe firmware (controller or namespace).
131 (0x83)	NVMe Feature 83h, Reservation Persistence; not supported by Ultrastar NVMe firmware (controller or namespace).

Example

```
hdm get-feature --feature 7 --path /dev/nvme0
```

Sample Output

```
[/dev/nvme0]
Device Type   = Ultrastar SN150
Device Path   = /dev/nvme0
UID           = 1C58STM0001A6758HUSPR3216AHP301
Feature Id    = 7
Feature Info  =
00000000 7f00 7f00 [.... ]
```

```
Results for get-feature: Operation succeeded.
```

get-info

The `get-info` command will retrieve the static properties of a drive, such as the device type, device path, UID, model name, serial number, namespace information, etc. associated with the device. The command can be used to extract information for a device or a namespace.

Synopsis

```
usage: hdm get-info          [--output-format FORMAT] [--config PATH]
                             [--path PATH |--uid UID |--alias ALIAS]
```

Retrieves properties from devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.

Examples

To extract information for a device:

```
hdm get-info --path /dev/nvme0
```

To extract information for a namespace:

```
hdm get-info --path /dev/nvme0n1
```

Device Sample Output

```
hdm get-info --path /dev/nvme0 --output-format mini

[/dev/nvme0]
Device Type           = HGST Ultrastar SN150
Device Path           = /dev/nvme0
UID                   = 1C58STM000198437HUSPR3232AHP301
Alias                  = sn15032TB
Vendor Name           = HGST
Model Name            = HUSPR3232AHP301
Serial Number         = STM0001994D7
Vendor ID             = 1C58
Firmware Version      = KMGNP698
Default Capacity      = 3200631791616
Namespace Count       = 4
Dual Port Support     = No
MPIO Support          = No
Namespace Sharing Support = No
Encryption Support    = No
```

Results for get-info: Operation succeeded.

Namespace Sample Output

```
hdm get-info --path /dev/nvme0n1 --output-format mini

[/dev/nvme0n1]
Device Type           = NVMe Namespace
Device Path           = /dev/nvme0n1
UID                   = 1C58STM000198437HUSPR3232AHP301
Alias                  = Namespace1
Parent UID            = 1C58STM000198437HUSPR3232AHP301
Namespace ID          = 1
Size                  = 3200631791616
Sector Count          = 1638723477307392
Sector Size           = 512
Metadata Size         = 8
DIF Level             = 1
Private               = Yes
```

Results for get-info: Operation succeeded.

get-log-page

The `get-log-page` command will retrieve a log page as specified by the `--page` option. The `PAGE` value must either be an integer or the hexadecimal equivalent of the log page.

Synopsis

```
usage: hdm get-log-page      [--output-format FORMAT] [--config PATH]
                             [--path PATH | --uid UID | --alias ALIAS] --page PAGE
```

Retrieves log pages from devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--page PAGE</code>	Identifier of the page. This value must be either an integer (e.g. 1) or a prefixed hexadecimal number (e.g. 0xC1).

Options

--page Value	Description
1 (0x01)	NVMe Log Page 01h, Error Information; supported for Ultrastar NVMe controllers only.
2 (0x02)	NVMe Log Page 02h, SMART / Health Information. This is a raw dump of Log Page 2 (see the Sample Output); however the same information will appear in a detailed, parsed format when using the <code>get-smart</code> command.
3 (0x03)	NVMe Log Page 03h, Firmware Information; supported for Ultrastar NVMe controllers only. This is a raw dump of Log Page 3 (see the Sample Output); however the same information will appear in a detailed, parsed format when using the <code>get-info</code> command.
193 (0xC1)	NVMe Log Page C1h, Vendor-Unique Log Page C1h; supported for Ultrastar NVMe controllers only. This is a raw dump of Log Page C1h (0xC1) (see the Sample Output).

Example

```
hdm get-log-page --page 0x02 --path /dev/nvme0
```

OR

```
hdm get-log-page --page 2 --path /dev/nvme0
```

Sample Output

```
[/dev/nvme0]
Device Type = Ultrastar SN150
Device Path = /dev/nvme0
UID         = 1C58STM0001A6758HUSPR3216AHP301
Page ID     = 2
Page Data   =
00000000  0038 0164 0a00 0000  0000 0000 0000 0000  [.8.d.....]
00000010  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000020  0700 0000 0000 0000  0000 0000 0000 0000  [.....]
00000030  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000040  0103 0000 0000 0000  0000 0000 0000 0000  [.....]
00000050  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000060  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000070  0100 0000 0000 0000  0000 0000 0000 0000  [.....]
00000080  0100 0000 0000 0000  0000 0000 0000 0000  [.....]
00000090  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000000A0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000000B0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000000C0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000000D0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000000E0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000000F0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000100  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000110  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000120  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000130  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000140  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000150  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000160  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000170  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000180  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
00000190  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000001A0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000001B0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000001C0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000001D0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000001E0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
000001F0  0000 0000 0000 0000  0000 0000 0000 0000  [.....]
```

Results for get-log-page: Operation succeeded.

get-smart

The `get-smart` command will retrieve the S.M.A.R.T. statistics and threshold values associated with the target device or namespace. As shown in the [Sample Output](#), the `get state` command will return error rates, read/write rates, critical temperature data, and the current properties of the NAND media.

Synopsis

```
usage: hdm get-smart          [--output-format FORMAT] [--config PATH]
                               [--path PATH | --uid UID | --alias ALIAS]
```

Retrieves SMART statistics and thresholds from devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.

Examples

To retrieve the SMART statistics and thresholds for a device:

```
hdm get-smart --path /dev/nvme0
```

To retrieve the SMART statistics and thresholds for a namespace:

```
hdm get-smart --path /dev/nvme0n1
```

Device Sample Output

```
hdm get-smart --path /dev/nvme0 --output-format mini

[/dev/nvme0]
Device Type                = Ultrastar SN150
Device Path                = /dev/nvme0
UID                        = 1C58STM0001A82F8HUSPR3216AHP301
Read Error Rate            = 4.863308
Write Error Rate           = 0.000000
Erase Error Rate           = 0.000000
Verify Error Rate          = 0.000000
Data Units Read            = 55557329
Data Units Written         = 115373711
Host Read Commands         = 3019544768
Host Write Commands        = 5609101100
Available Spare             = 100
Available Spare Threshold  = 10
Available Spare Critical Warning = false
Percentage Used            = 0
Temperature                = 45
Temperature Threshold      = 90
Temperature Critical Warning = false
Reference Temperature      = 90
Maximum Temperature        = 73
Minimum Temperature        = 24
Total Time Over Reference Temperature = 0
Power On Hour When At Max Temperature = 2858
Power On Hour When At Min Temperature = 7
Device Reliability Critical Warning = false
Media Read Only Mode Critical Warning = false
Volatile Memory Backup Device Failure Critical Warning = false
Controller Busy Time       = 1855
Power Cycles               = 2
Power On Hours             = 216
Unsafe Shutdowns           = 1
Media And Data Integrity Errors = 0
Number Of Error Information Log Entries = 1
Results for get-smart: Operation succeeded.
```

Namespace Sample Output

```
hdm get-smart --path /dev/nvme0n1 --output-format mini

[/dev/nvme0n1]
Device Type                = NVMe Namespace
Device Path                = /dev/nvme0n1
UID                        = 000CCA006005D100
Data Units Read            = 55557329
Data Units Written         = 115373711
Host Read Commands         = 3019544768
Host Write Commands        = 5609101100
Available Spare             = 100
Available Spare Threshold  = 10
Available Spare Critical Warning = false
Percentage Used            = 0
Temperature                = 45
Temperature Critical Warning = false
Device Reliability Critical Warning = false
Media Read Only Mode Critical Warning = false
Volatile Memory Backup Device Failure Critical Warning = false
Controller Busy Time       = 1855
Power Cycles               = 2
Power On Hours             = 216
Unsafe Shutdowns           = 1
Media And Data Integrity Errors = 0
Number Of Error Information Log Entries = 0
Results for get-smart: Operation succeeded.
```

get-state

The `get-state` command will retrieve the current state information of the target device or namespace, including the current encryption mode and remaining life.

Synopsis

```
usage: hdm get-state          [--output-format FORMAT] [--config PATH]
                               [--path PATH |--uid UID |--alias ALIAS]
```

Retrieves state information from devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.

Examples

To retrieve the current state information for a device:

```
hdm get-state --path /dev/nvme0
```

To retrieve the current state information for a namespace:

```
hdm get-state --path /dev/nvme0n1
```

Device Sample Output

```
hdm get-state --path /dev/nvme0 --output-format mini

[/dev/nvme0]
Device Type      = Ultrastar SN150
Device Path      = /dev/nvme0
UID              = 1C58STM0001A82F8HUSPR3216AHP301
Device Status    = Ready
Power Consumption = 6.914188
Life Gauge       = 100
Dual Port Mode   = Single Port
Encryption Mode  = No Encryption
```

Results for get-state: Operation succeeded.

Namespace Sample Output

```
hdm get-state --path /dev/nvme0n1 --output-format mini

[/dev/nvme0n1]
Device Type      = NVMe Namespace
Device Path      = /dev/nvme0n1
UID              = 000CCA006005D100
Life Gauge       = 100
```

Results for get-state: Operation succeeded.

get-statistics

The `get-statistics` command retrieves statistical data according to the specified `-category`. The set of valid intervals as specified by the `--interval` may vary by device type and only apply to the `performance` category; the device stores various statistical fields that are used to review and analyze the device performance. There are fifteen sets of performance statistics (1-15) that can be collected at various intervals. If an `--interval` is not specified for the `performance` category, then the value that corresponds to the lifetime statistics of the device is used as the default. The command is only supported for physical devices.

Synopsis

```
usage: hdm get-statistics  [--output-format FORMAT] [--config PATH]
                           [--path PATH |--uid UID |--alias ALIAS]
                           [--interval INTERVAL] [--category CATEGORY]
```

Retrieves statistics from devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--interval INTERVAL</code>	Interval to be used for this operation.
<code>--category CATEGORY</code>	Category to be used for this operation.

Options

Option	Value	Description
<code>--interval INTERVAL</code>	1	Most recent five (5) minute accumulated set.
	2 - 12	Previous five (5) minute accumulated sets.
	13	The accumulated total of sets 1 through 12 that contain the previous hour of accumulated statistics.
	14	The statistical set accumulated since power-up.
	15	The statistical set accumulated during the entire lifetime of the device.
<code>--category CATEGORY</code>	performance	Performance statistics; the properties and values collected by this category will contain, if applicable, the properties and values defined in the Performance Log Page (37h) (SCSI).
	power	Power statistics; the properties and values collected by this category will contain power consumption statistics. The power consumption value is in mW.
	queues	Queue statistics; the properties and values collected by this category are applicable to queues that are used by the device driver or firmware.
	commands	Command statistics; the properties and values collected by this category will contain read, write, verify and erase command execution statistics.
	miscellaneous	Miscellaneous statistics; the properties and values collected by this category are not specific to any of the previous categories.

Example

```
hdm get-statistics --interval 14 --category performance --path /dev/nvme0
```

Sample Output --category performance

```
[/dev/nvme0]
Device Type           = Ultrastar SN100
Device Path           = /dev/nvme0
UID                   = 1C58STM0001A753DHUSPR3216ADP301
Category              = Performance
Interval              = 15
Host Read Commands    = 160388108
Host Read Blocks      = 41058717074
Average Read Size     = 255.996019
Host Read Cache Hit Commands = 33
Host Read Cache Hit Percentage = 0
Host Read Cache Hit Blocks = 8448
Average Read Cache Hit Size = 256.000000
Host Read Commands Stalled = 13
Host Read Commands Stalled Percentage = 0
Host Write Commands   = 177210509
Host Write Blocks     = 45365897868
Average Write Size    = 256.000043
Host Write Odd Start Commands = 26538833
Host Write Odd Start Commands Percentage = 14
Host Write Odd End Commands = 26538833
Host Write Odd End Commands Percentage = 14
Host Write Commands Stalled = 2512659
Host Write Commands Stalled Percentage = 1
NAND Read Commands    = 6574808166
NAND Read Blocks      = 210393861312
Average NAND Read Size = 32.000000
NAND Write Commands   = 2109949049
NAND Write Blocks     = 67518369568
Average NAND Write Size = 32.000000
NAND Read Before Write = 56049544
```

Results for get-statistics: Operation succeeded.

Sample Output --category power

```
[/dev/nvme0]
Device Type           = Ultrastar SN100
Device Path           = /dev/nvme0
UID                   = 1C58STM0001A753DHUSPR3216ADP301
Category              = Power
Processor Power Consumption = 7503
NAND Power Consumption   = 2
```

Results for get-statistics: Operation succeeded.

Sample Output --category queues

```
[/dev/nvme0]
Device Type           = Ultrastar SN100
Device Path           = /dev/nvme0
UID                   = 1C58STM0001A753DHUSPR3216ADP301
Category              = Queues
Number of I/O Completion Queues Allocated = 128
Number of I/O Submission Queues Allocated = 128
```

Results for get-statistics: Operation succeeded.

Sample Output --category commands

```
[/dev/nvme0]
Device Type           = Ultrastar SN100
Device Path           = /dev/nvme0
UID                   = 1C58STM0001A753DHUSPR3216ADP301
Category              = Commands
Total Re-Writes       = 0
Total Write Bytes Processed = 518332416
Total Uncorrected Write Errors = 0
Flash Write Commands  = 7336
Read Errors Corrected Without Delays = 787
Read Errors Corrected With Possible Delays = 0
Total Re-Reads        = 0
Total Read Errors Corrected = 787
Total Read Bytes Processed = 0
Total Uncorrected Read Errors = 0
Flash Read Commands    = 888
Total XOR Recovered    = 0
Total Corrected Bit Count = 787
Total Uncorrected Erase Errors = 0
Flash Erase Commands   = 311
Manufacturers Defect Count = 0
Grown Defect Count     = 0
Maximum Erase Count of User Data = 19
Maximum Erase Count of System Data = 0
```

Results for get-statistics: Operation succeeded.

Sample Output --category miscellaneous

```
[/dev/nvme0]
Device Type           = Ultrastar SN100
Device Path           = /dev/nvme0
UID                   = 1C58STM0001A753DHUSPR3216ADP301
Category              = Miscellaneous
Firmware Download Number = 4
PFA Counter           = 0
```

Results for get-statistics: Operation succeeded.

locate

The `locate` command will either activate or deactivate the beacon LED on the device(s), allowing the user to find the physical device(s) installed in the host system. The user must specify the [device-reference](#) for the target device. The `--enable` option is the default action that activates the beacon if it is inactive. The user specifies the `--disable` option to deactivate the beacon if it is currently active. The command is only valid for physical devices.

Synopsis

```
usage: hdm locate          [--output-format FORMAT] [--config PATH]
                           [--path PATH |--uid UID |--alias ALIAS]
                           [--disable] [--enable]
```

Enables or disables the LED beacon for physical devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--disable</code>	Executes the disable action.
<code>--enable</code>	Executes the enable action.

Examples

To enable the beacon LED for a device:

```
hdm locate --enable --path /dev/nvme0
```

To disable the beacon LED for a device:

```
hdm locate --disable --path /dev/nvme0
```

manage-firmware

The `manage-firmware` command will list the firmware version(s) present on the device; the user can also perform a partial or full update of the firmware on the device.

Synopsis

```
usage: hdm manage-firmware  [--output-format FORMAT] [--config PATH]
                             [--path PATH |--uid UID |--alias ALIAS]
                             [--list] [--load] [--file FILE] [--activate]
                             [--slot NUMBER] [--reset]
```

Displays and updates firmware on physical devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--list</code>	Display information from the operation.
<code>--load</code>	Load a firmware image onto the specified devices.
<code>--file FILE</code>	File name or path.
<code>--activate</code>	Activate a firmware image.
<code>--slot NUMBER</code>	The slot number.
<code>--reset</code>	Reset the specific devices.

Options

Option	Description
<code>--list</code>	The <code>--list</code> switch will perform the default action and display information about the firmware image(s) if no other options are specified.
<code>--load</code>	The <code>--load</code> switch will install the firmware, as specified by the <code>--file</code> parameter, onto the target device(s).
<code>--file FILE</code>	The <code>FILE</code> value is the path and filename of the image. The <code>--file FILE</code> option is required with the <code>--load</code> switch.
<code>--activate</code>	The <code>--activate</code> switch is used to enable the firmware on the device as per the required <code>--slot NUMBER</code> option. This switch may be used along with the <code>--load</code> switch to immediately activate the firmware upon the successful load.
<code>--slot NUMBER</code>	The <code>--slot</code> switch specifies the slot number to target or devices that support multiple firmware images. This switch is required with the <code>--load</code> and <code>--activate</code> switches. The user should be aware that Slot 1 is read-only.
<code>--reset</code>	The <code>--reset</code> switch is used to perform a bus reset on the slot (<code>--slot NUMBER</code>) and indicates that the <code>--activate</code> switch should automatically perform a controller/bus reset after completing the activation. The switch is only valid when used with the <code>--activate</code> switch.

Examples

To list information about the firmware image(s):

```
hdm manage-firmware --list --path /dev/nvme0
```

```
[/dev/nvme0]
Device Type = Ultrastar SN100
Device Path = /dev/nvme0
UID         = 1C58STM0001A753DHUSPR3216ADP301
Slot 1      = KMGNP69E Loaded (Read-only)
Slot 2      = KMGNP69A Pending
Slot 3      = KMGNP69E Loaded
Slot 4      = KMGNP69A Active
```

Results for manage-firmware: Operation succeeded.

To load a firmware image onto a device:

```
hdm manage-firmware --load --file /firmware/filename.bin --slot 2 --path /dev/nvme0
```

To activate a firmware image loaded onto the device:

```
hdm manage-firmware --activate --slot 2 --path /dev/nvme0
```

To perform a controller / bus reset after activating the firmware on a device:

```
hdm manage-firmware --activate --reset --slot 2 --path /dev/nvme0
```

Firmware Slot States

The following is for informational purposes only. The firmware loaded on a particular slot (Slot 1, Slot 2, etc.) may be in one of three states: Active, Loaded or Pending.

Slot State	Description
Active	The Active state indicates that the currently loaded firmware is running.
Loaded	The Loaded state indicates that the firmware has been loaded on the slot, but is not yet activated (Active).
Pending	The Pending state indicates that the firmware has been Loaded and will become Active following the next drive reset or system power-cycle.

manage-namespaces

The `manage-namespaces` command allows the user to perform interrelated namespace management tasks. The user can target multiple namespaces by specifying multiple `--id namespace-id` options in the command syntax. The *device-reference* must refer to a physical device. The user will be prompted to confirm unless the `--force` option is specified.

Synopsis

```
usage: hdm manage-namespaces  [--output-format FORMAT] [--config PATH]
                               [--path PATH |--uid UID |--alias ALIAS]
                               [--list] [--create] [--attach] [--detach]
                               [--delete] [--size SIZE] [--id ID] [--force]
```

Displays and manages namespaces on a capable physical device.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--list</code>	Display information from the operation.
<code>--create</code>	Executes the create action.
<code>--attach</code>	Executes the attach action.
<code>--detach</code>	Executes the detach action.
<code>--delete</code>	Executes the delete action.
<code>--size SIZE</code>	Size in gigabytes.
<code>--id ID</code>	The identifier to be used for this operation.
<code>--force</code>	Bypass the confirmation messages.

Options

Option	Description
<code>--list</code>	The <code>--list</code> option will present information regarding the namespace(s) currently present on the device(s). This is the default action if none of the remaining options are specified.
<code>--create</code>	This option will create a new namespace or namespaces on the target device. The <code>--size</code> option is required for this option. As an option, the ID integer can be specified using the <code>--id</code> option. If no ID integer is specified, then a single namespace will be created having the lowest available namespace identifier. The namespace will be formatted with the default LBA format and DIF level (for Ultrastar NVMe controllers this is 512-bytes sector size, 0 metadata size and a DIF level of 0 (zero)).
<code>--attach</code>	This option will attach (if possible) the namespace specified by the <code>--id</code> option integer.
<code>--detach</code>	This option will detach the namespace specified by the <code>--id</code> option integer.
<code>--delete</code>	This option will delete the namespace specified by the <code>--id</code> option integer. A permanent namespace cannot be deleted. The namespace is automatically detached before deletion.
<code>--size SIZE</code>	This option will set, in gigabytes, the size of the namespace. This option is only valid when used with the <code>--create</code> option.
<code>--id ID</code>	The <code>--id</code> option allows the user to target an existing namespace or multiple namespaces using the actual ID integer value created by the user, e.g., <code>--id 1</code> , <code>--id 2</code> , etc.
<code>--force</code>	The <code>--force</code> option will perform the operation without any user confirmation.

Examples

To list all the current namespace assignments:

```
hdm manage-namespaces --list --path /dev/nvme0 --output-format mini
```

OR

```
hdm manage-namespaces --path /dev/nvme0 --output-format mini
```

```
[/dev/nvme0]
Device Type   = Ultrastar SN150
Device Path   = /dev/nvme0
UID           = 1C58STM00019EE74HUSPR1616AHP301
Alias         = ctrlr
Namespace 1   = Attached 600 GB
Namespace 3   = Detached
Namespace 4   = Attached 100 GB
```

Results for manage-namespaces: Operation succeeded.

To create a new namespace on the target device:

```
hdm manage-namespaces --create --id 3 --size 1600 --path /dev/nvme0
```

To delete the namespace on the target device:

```
hdm manage-namespaces --delete --id 3
```

To attach the namespace:

```
hdm manage-namespaces --attach --id 3
```

To detach the namespace:

```
hdm manage-namespaces -detach --id 3
```

manage-power

The `manage-power` command will list the available power states for a device and allows the user to apply a specific power state to a device. The [device-reference](#) must be a physical device, as namespaces are not valid.



Ultrastar NVMe PCIe Devices Only: Issuing some HDM commands, or performing a host reboot/power cycle after using the `manage-power` command to change the power state will revert the power state to its previous value. The user should therefore perform and complete all other device tasks before attempting to alter the power state using the `manage-power` command.

Synopsis

```
usage: hdm manage-power          [--output-format FORMAT] [--config PATH]
                                [--path PATH |--uid UID |--alias ALIAS]
                                [--list] [--change] [--state STATE]
```

Displays and controls power states of physical devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--list</code>	Display information from the option.
<code>--change</code>	Executes the change action.
<code>--state STATE</code>	The state to be used for the action.

Options

Option	Description
<code>--list</code>	The <code>--list</code> option will present information regarding the power states that are available for the device. This is the default action if none of the remaining options are specified.
<code>--change</code>	This option will change and apply the power state on the targeted device.
<code>--state STATE</code>	This option specifies the power state that should be applied to the targeted device; the option is only valid when used with the <code>--change</code> option. The valid values for the power-state are those listed with the prefix of "Power State".

Examples

To list the available power states for a device:

```
hdm manage-power --list --path /dev/nvme0 --output-format mini
```

OR

```
hdm manage-power --path /dev/nvme0 --output-format mini
```

```
[/dev/nvme0]
Device Type   = Ultrastar SN150
Device Path   = /dev/nvme0
UID           = 1C58STM00019EE74HUSPR1616AHP301
Alias         = sn1004TB
Power State 1 = 25 W (Active)
Power State 2 = 20 W
Power State 3 = 15 W
Power State 4 = 10 W
Power State 5 = 10 W [no I/O]
```

```
Results for manage-power: Operation succeeded.
```

To apply a valid power state to a device:

```
hdm manage-power --path /dev/nvme0 --change --state 3
```

monitor

The `monitor` command will retrieve device information, state, SMART and statistics for all supported devices in the system, or for a specific device according to the [device-reference](#). The data is retrieved and refreshed by the value specified by the `--interval` option. The command will continue to run until it is killed by the user or until the number of `--count` iterations has been completed.



The user should be aware that while there are no minimum or maximum `--interval` or `--count` values, no extreme edge cases have been attempted. Users are therefore advised to use discretion when specifying the values for these options.

Synopsis

```
usage: hdm monitor          [--output-format FORMAT] [--config PATH]
                             [--path PATH |--uid UID |--alias ALIAS]
                             --interval INTERVAL [--count NUM]
```

Continuously retrieves device information at a specified interval.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--interval INTERVAL</code>	Interval to be used for this operation.
<code>--count NUM</code>	The number of times to execute the operation.

Options

Option	Description
<code>--interval INTERVAL</code>	The <code>INTERVAL</code> value is expressed in seconds, e.g., <code>--interval 30</code> would specify an interval of thirty (30) seconds between the logging each device event.
<code>--count NUM</code>	The <code>NUM</code> value expresses the number of iterations to complete before ending the process.

Examples

To monitor a specific device at intervals of 30 seconds:

```
hdm monitor --interval 30 --path /dev/nvme0
```

To monitor all devices at an interval of 30 seconds:

```
hdm monitor --interval 30
```

To monitor a specific device at intervals of 30 seconds but limit the polling to 10 iterations:

```
hdm monitor --interval 30 --count 10 --path /dev/nvme0
```

To monitor all devices at an interval of 30 seconds but limit the polling to 10 iterations:

```
hdm monitor --interval 30 --count 10
```

prepare-for-removal

The `prepare-for-removal` command will instruct the operating system to stop issuing I/O to the target device; the device may then be safely removed from the system. The user must reboot the host or power-cycle the system before the OS will redetect the device. This command is only valid for physical devices.

Synopsis

```
usage: hdm prepare-for-removal    [--output-format FORMAT] [--config PATH]
                                   [--path PATH | --uid UID | --alias ALIAS]
```

Prepares physical devices for safe removal from the system.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.

Example

```
hdm prepare-for-removal --path /dev/nvme0
```

reset-to-defaults

The `reset-to-defaults` command will restore the targeted device(s) to the factory defaults, according to the device type, model, serial number and SMART thresholds; however, neither the firmware nor the statistical reporting is affected. All user data, including namespaces and partitions will be destroyed. The command is only valid for physical devices. The user will be prompted to confirm unless the `--force` option is specified.

Synopsis

```
usage: hdm reset-to-defaults      [--output-format FORMAT] [--config PATH]
                                   [--path PATH | --uid UID | --alias ALIAS] [--force]
```

Resets physical devices to factory default configuration.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--force</code>	Bypass all confirmation messages.

Examples

To reset a device to the defaults with a confirmation prompt:

```
hdm reset-to-defaults --path /dev/nvme0
```

To reset a device to the defaults without a confirmation prompt:

```
hdm reset-to-defaults --force --path /dev/nvme0
```

resize

The `resize` command will set the user capacity of a device according to the value of the `--size` parameter. *The user should be aware that all user data, namespaces and partitions will be destroyed.* The valid value for the `--size` parameter is dependent upon the device model and must be expressed in gigabytes (GB). For example, if the device is 2.0TB (2000GB), but the user wants to resize it to 1.6TB (1600GB), the expression would be `--size 1600`. The command is only valid for physical devices. The user will be prompted to confirm unless the `--force` option is specified.

Synopsis

```
usage: hdm resize          [--output-format FORMAT] [--config PATH]
                          [--path PATH |--uid UID |--alias ALIAS]
                          [--size SIZE] [--force]
```

Changes the user capacity of physical devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--size SIZE</code>	Size in gigabytes.
<code>--force</code>	Suppresses the confirmation prompt.

Note: The value for the `--size SIZE` must fall within the following ranges for Ultrastar NVMe controllers as listed in the following table:

Capacity	Minimum Resize Value (GB)	DWD	Maximum Resize Value (GB)	DWD
800GB	560	7.4	955	1.1
1.6TB	1120	7.5	1910	1.2
3.2TB	2240	7.5	3820	1.2

Examples

To resize a device with a confirmation prompt:

```
hdm resize --size 1600 --path /dev/nvme0
```

To resize a device without a confirmation prompt:

```
hdm resize --force --size 1600 --path /dev/nvme0
```

scan

The `scan` command will retrieve all controllers and namespaces attached to the host and presents them in a formatted list.

Synopsis

```
usage: hdm scan          [--output-format FORMAT] [--config PATH]
```

List the storage devices attached to this host.

OPTIONS:

```
--output-format FORMAT  Output format for the command. Valid output formats are
                        "text", "mini", "csv", and "json". The default is "mini".
--config PATH           Path to the HDM configuration file.
```

Example

```
hdm scan
```

Sample Output

```
[1C58STM0001A6758HUSPR3216AHP301]
Device Type      = Ultrastar SN150
Device Path      = /dev/nvme0
UID              = 1C58STM0001A6758HUSPR3216AHP301
Vendor Id        = 1C58
Device Id        = 0003
Vendor Name      = HGST
Model Name       = HUSPR3216AHP301

[000CCA0060032E80]
Device Type      = NVMe Namespace
Device Path      = /dev/nvme0n1
UID              = 000CCA0060032E80
Parent UID       = 1C58STM0001A6758HUSPR3216AHP301
Parent Device Path = /dev/nvme0
Namespace ID     = 1
```

Results for scan: Operation succeeded on 2 of 2 devices.

secure-erase

The `secure-erase` command will perform a secure erase of the target device using an erase level specified by the user. The command applies the erase to the entire device and cannot target a specific namespace or partition. The operation will result in a reset of the LBA format to the default for the targeted device (for Ultrastar NVMe devices, the default LBA format is 512-byte sectors with no metadata or T10 DIF Protection). The user will be prompted to confirm unless the `--force` option is specified.

Synopsis

```
usage: hdm secure-erase    [--output-format FORMAT] [--config PATH]
                           [--path PATH | --uid UID | --alias ALIAS]
                           [--type ERASE_TYPE] [--force]
```

Securely erases all user data from physical devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--type ERASE_TYPE</code>	Type of this operation to perform.
<code>--force</code>	Bypass all confirmation messages.

Options

Option	Value	Description
<code>--type ERASE_TYPE</code>	user	If the <code>ERASE_TYPE</code> = user, then the command erases all user content present in the NVM subsystem.
	cryptographic	If the <code>ERASE_TYPE</code> = cryptographic, then the command will erase all content present in the NVM subsystem by deleting the encryption key with which the user data was previously encrypted. Note: The user should be aware that not all devices support the <code>cryptographic</code> option.

Examples

To perform a secure-erase of all the user data with a confirmation prompt:

```
hdm secure-erase --type user --path /dev/nvme0
```

To perform a cryptographic secure-erase with a confirmation prompt:

```
hdm secure-erase --type cryptographic --path /dev/nvme0
```

To perform a secure-erase of all the user data without a confirmation prompt:

```
hdm secure-erase --force --type user --path /dev/nvme0
```

To perform a cryptographic secure-erase without a confirmation prompt:

```
hdm secure-erase --force --type cryptographic --path /dev/nvme0
```

secure-purge

The `secure-purge` command will perform a purge on a physical device. Whereas `secure-erase` only erases the user data areas, `secure-purge` applies a sanitize/erase fill to the entire device and does not target a specific namespace or partition. The process will erase all the data (user and system areas) and then fills (overwrites) each addressable block of memory with a predetermined pattern. The user will be prompted to confirm unless the `--force` option is specified.

Synopsis

```
usage: hdm secure-purge          [--output-format FORMAT] [--config PATH]
                                [--path PATH | --uid UID | --alias ALIAS] [--force]
```

Performs a secure purge on physical devices.

OPTIONS:

<code>--output-format FORMAT</code>	Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".
<code>--config PATH</code>	Path to the HDM configuration file.
<code>--uid UNIQUE_ID</code>	Device unique identifier.
<code>--path DEV_PATH</code>	Platform path of the device as shown in the output of <code>hdm scan</code> .
<code>--alias ALIAS</code>	User supplied name for the device.
<code>--force</code>	Bypasses all confirmation messages.

Examples

To perform a purge of the device with a confirmation prompt:

```
hdm secure-purge --path /dev/nvme0
```

To perform a purge of the device without a confirmation prompt:

```
hdm secure-purge --force --path /dev/nvme0
```

version

The `version` command will obtain and display the current build and version of the software. The output consists of a string that displays the current software version.

Synopsis

```
usage: hdm version          [--output-format FORMAT] [--config PATH]
```

Display HDM build and version information.

OPTIONS:

`--output-format FORMAT` Output format for the command. Valid output formats are "text", "mini", "csv", and "json". The default is "mini".

`--config PATH` Path to the HDM configuration file.

Example

```
hdm version
```

Output

```
HDM Version = 3.1.0
Build Date  = 2015-07-07T14:20:30
Build Host  = hdmtest-2
Commit ID   = 16c7d77a1c48334950265dd68fbcf3aee36767a4
Ref Spec    = refs/heads/3.1_release
```

Results for version: Operation succeeded.

Glossary

Table 7: Glossary of Terms

Term	Definition
CLI	Command Line Interface
CSV	Comma-Separated Values or Character-Separated Values.
EULA	End-User License Agreement
FAE	Field Application Engineer
Fibre Channel	This term refers to devices that use the Fibre Channel-Arbitrated Loop (FC-AL) interface. It is a networking technology and protocol primarily used for storage networking.
Firmware	Firmware is a term often used to denote the embedded software programs and/or data structures used to internally control electronic devices, such as microcontrollers or microprocessors.
FW	See <i>Firmware</i> .
GNU	A recursive acronym for “ <i>GNU's Not Unix!</i> ”
GPL	GNU Public License.
HBA	Host Bus Adapter
HDD	Hard Disk Drive
IO or I/O	Input/Output. A READ or WRITE operation of user data.
IOPS	Input/Output Per Second, usually measured in 4,096 byte sizes.
JSON	JavaScript Object Notation. JSON is an open standard format that uses human-readable text to transmit data objects consisting of attribute–value pairs and is used to transmit data between a server and web application.
Operating System	An Operating System (OS), also known as system software, manages all interaction and services between the user, applications, and system resources (hardware).
OS	See <i>Operating System</i> .
PCIe	PCI Express, or Peripheral Component Interconnect Express, is a computer expansion card standard. The term refers to add-in devices that use the PCIe serial interface.
SCSI	Small Computer System Interface
Serial-Attached SCSI	This term refers to devices that use the Serial-Attached SCSI, or SAS interface.
SMART (S.M.A.R.T.)	<i>Self-Monitoring, Analysis and Reporting Technology</i> (S.M.A.R.T. or SMART) is a monitoring system for computer storage devices (HDDs and SSDs) that detects and reports on various indicators of drive reliability to anticipate hardware failures.
SSA	Solid-State Accelerator
SSD	Solid-State Drive or Solid-State Device

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